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The truth table for AND Gate is

Input		Output
A	B	C
0	0	0
0	1	0
1	0	0
1	1	1

Table 2.7 Truth Table for AND Gate

2.9.2 OR Gate

The OR gate gets its name from its behaviour like the logical inclusive "OR". The output is "true" if either or both of the inputs are "true". If both inputs are "false" then the output is "false". In otherwords the output will be 1 if and only if one or both inputs are 1; otherwise, the output is 0. The logical symbol of the OR gate is

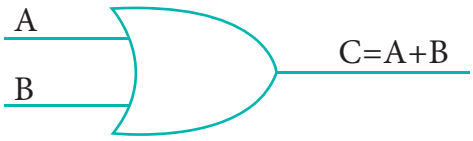

Fig. 2.5 Logic symbol of OR Gate

The OR gate output is

C = A OR B

We use the + sign to denote the OR function. Therefore,

C = A + B

Read this as "C equals A OR B".

For instance, if both the inputs are 1

C = A + B = 1 + 1 = 1

The truth table for OR gate is

Input		Output
A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

Table 2.8 Truth Table for OR Gate

2.9.3 NOT Gate

The NOT gate, called a logical inverter, has only one input. It reverses the logical state. In other words the output C is always the complement of the input. The logical symbol of the NOT gate is

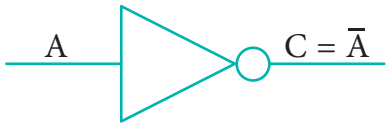

Fig. 2.6 Logic symbol of NOT Gate

The boolean function of NOT gate is

C = NOT A

In boolean algebra, the overbar stands for NOT operation. Therefore,

$C = \overline{A}$

Read this as "C equals NOT A" or "C equals the complement of A".

If A is 0,

$C = \overline{0} = 1$

On the otherhand, if A is 1,

$C = \overline{1} = 0$

The truth table for NOT gate is